ASHG 2026 · MONTRÉAL · BOARD [BOARD NUMBER]

SV-PGS: one Bayesian model for SNVs, indels, tandem repeats and structural variants in polygenic scores

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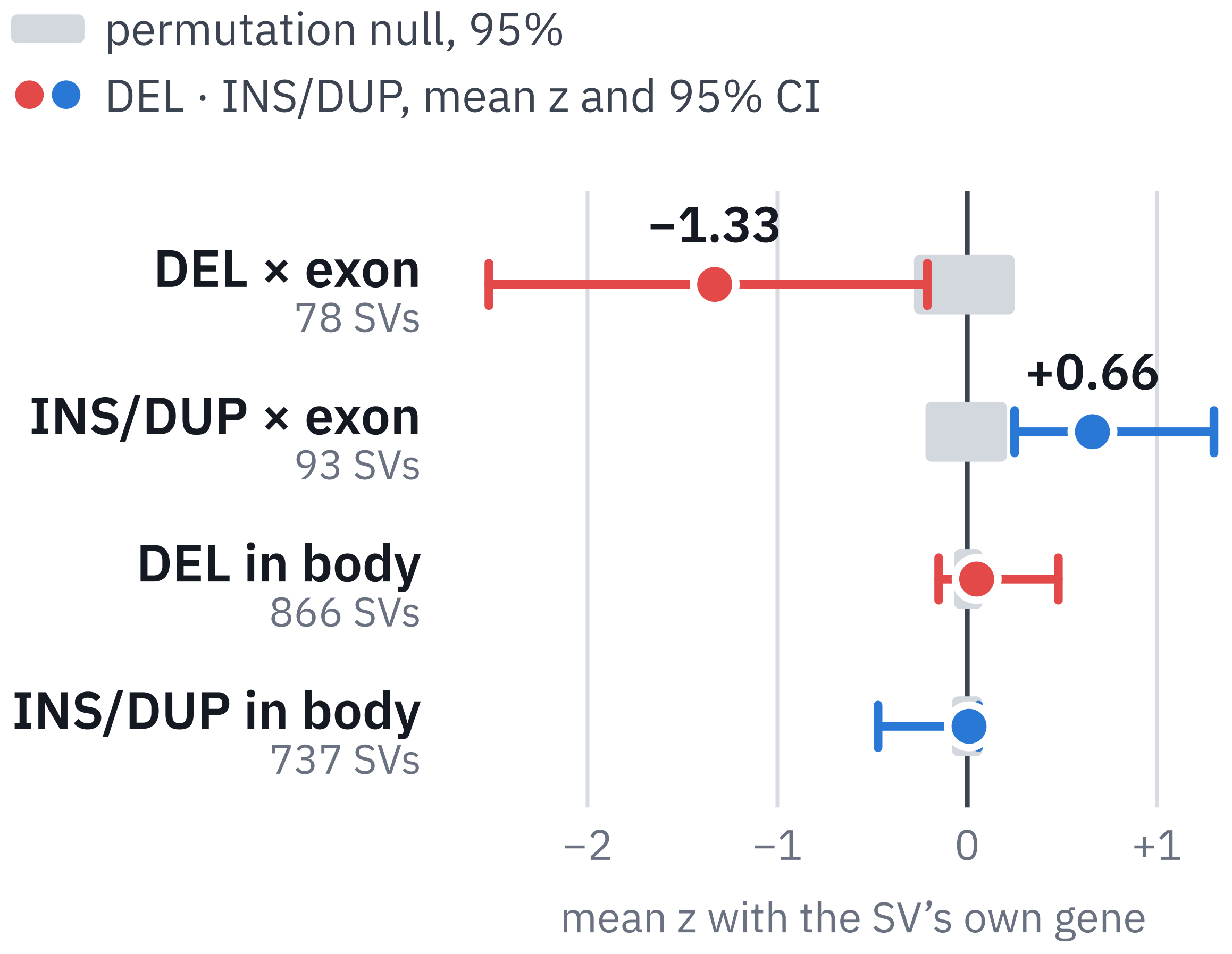
Methods and public-data results. SV-PGS accuracy and All of Us results are not yet available; no All of Us data appear here.

1 SV effects can have a sign

Polygenic scores mostly leave out structural variants, or credit them through tagging SNVs.

- SVs are enriched for heritability in **31 of 105** UK Biobank traits, median **14.8×** (MiXeR-SV, preprint).
- An SV's ALT allele is a physical event, so its effect can have a biological sign.

Exon deletions lower their gene's expression; exon insertions and duplications raise it `real`



MAGE RNA-seq with 1kGP SV calls: 534 people, 500 genes, 66,397 SVs. Exon rows: permutation p = 0.0005, the smallest 2,000 permutations give.

2 One model for every variant

Every record, from a single-base indel to a large deletion, enters one model. No length bins.

$$y = C\alpha + \Sigma_j D_j \beta_j + \varepsilon$$

$$\beta_j \sim \int \mathrm{N}(0, u_j s)\, g_c(s)\, \mathrm{d}s$$

$$\log u_j = \underbrace{\log(\kappa_j^2 \operatorname{Var} D_j)}_{\text{reliability}} + \underbrace{h(\log \operatorname{Var} G_j)}_{\text{frequency}} + \underbrace{d_j^\top \theta}_{\text{annotations}}$$

- **The mixing density g_c is learned per class**: continuous, nonparametric, roughness-penalized. Gaussian and BayesR-style priors are special cases.
- **Annotations are smooths** with learned smoothness: log length, distance to TSS, allele frequency.
- **A tandem-repeat locus is one column**: its signed total length.
- **Imputed SVs act like confident draws**, not calibrated means: at MAF 1–5%, κ 0.80 vs √r² 0.83 `semi-real`. Reliability sets each variant's prior scale.

3 Fit it; refuse what did not converge

Empirical Bayes: hyperparameters maximize the approximate evidence at a fixed point of the approximate posterior.

- Newton steps on the total curvature, with a trust region.
- **A fit is scored only if converged**: remaining gain ≤ 1/(2K) nats. Others are quarantined.

A flat prior on a smooth's linear term has no maximum

Empirical Bayes walked a coefficient to 5 × 10⁴. A learned ridge on each smooth's null space fixed it `machinery`:

sample route	**8 s**	~~was 50 s~~
LD-band route	**19 s**	~~was > 10 min~~

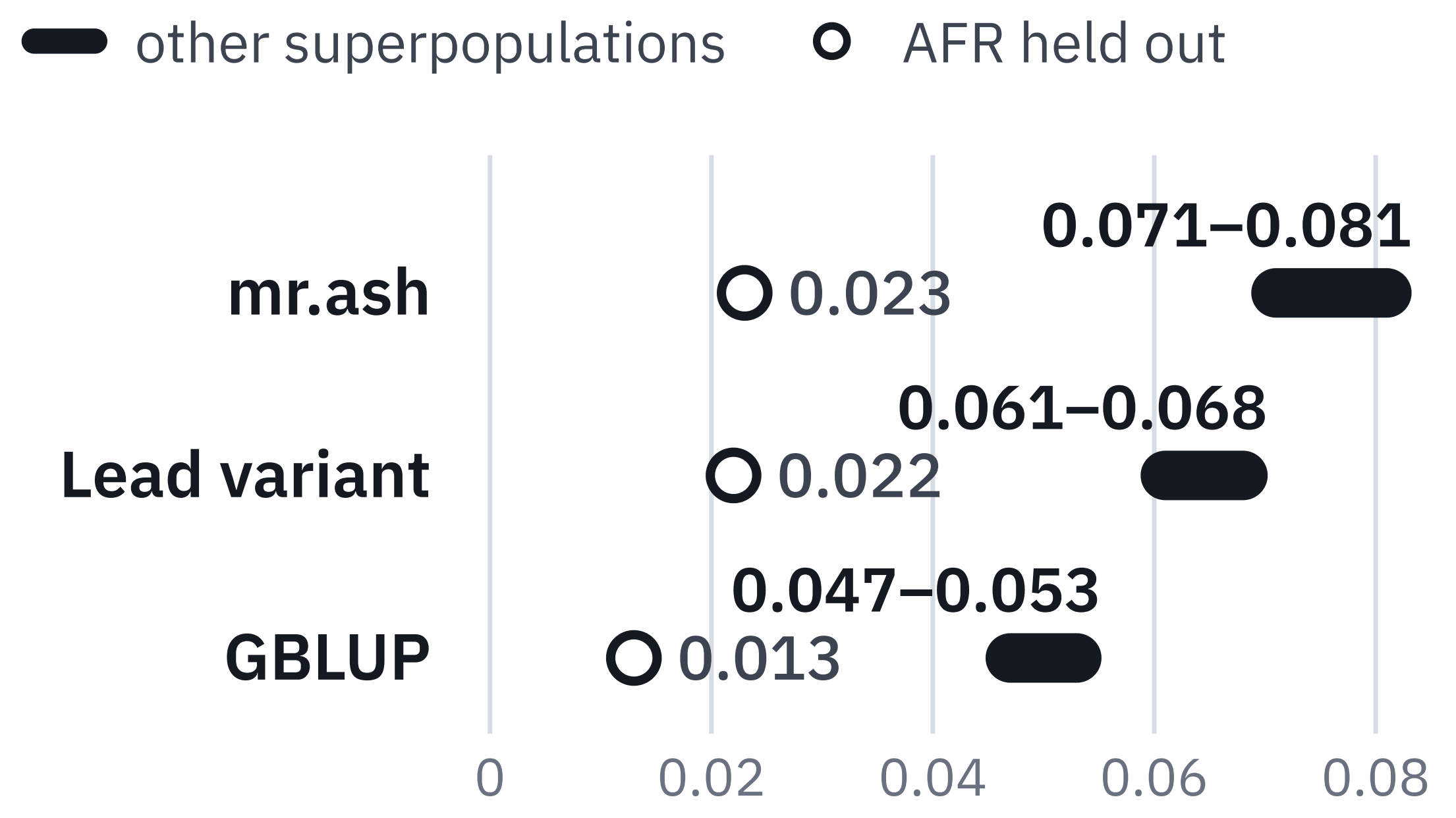
~100 s	to score 100k people × 17.3M variants on one H100
149–180 s	12-model solve, 30k × 484k, one A40
0.5–3.5%	marginal-variance error (block-Jacobi 16–58%)

Speed and exactness, against exact references `machinery`

4 Public benchmarks

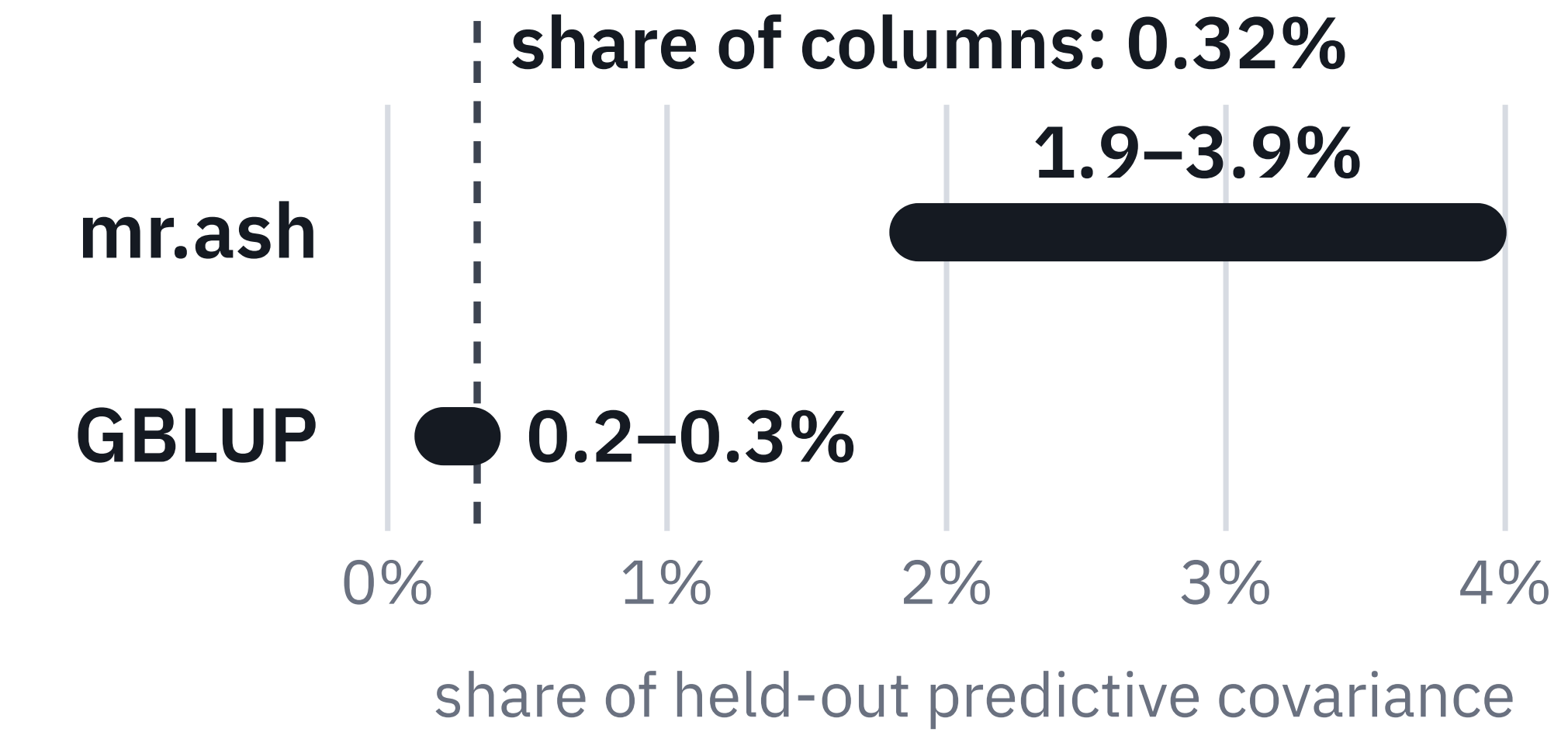
- **bench-real** `real`: MAGE expression, 730 people, 1kGP SNV, indel and SV calls.
- **bench-sim**: 50,000 1kGP mosaic genomes, SVs re-imputed, sealed seeds.

Baseline held-out r² on bench-real `real`



Leave-one-superpopulation-out. Holding out AFR costs about 70% of r².

A sparse prior gives SVs 6–12× their share of columns `real`



5 The All of Us study

About 100,000 multi-ancestry participants with imputed genomes. Set up; not yet run.

`snv_indel`	`with_sv`
records under 50 bp	every record + GATK-SV calls

- Same people, folds and covariates in both arms; 5 kinship-grouped folds.
- Endpoints: log loss, Brier score and AUC of calibrated probabilities.

hypertension · coronary artery disease · asthma · hyperlipidemia · breast cancer · type 2 diabetes · atrial fibrillation · hypothyroidism · COPD · depression · gout · cataract · CKD · psoriasis · prostate cancer

● **pipeline runs end to end on synthetic data**
○ GPU validation
○ full-genome pilot, one trait
○ 180 fits: 15 diseases × 2 arms × 6
○ held-out evaluation

No All of Us data or results appear on this poster.

EVIDENCE TAGS

`real`	real phenotypes and genotypes, nothing simulated
`semi-real`	real haplotypes and imputation, simulated effects
`machinery`	speed or exactness only; never an accuracy claim

REFERENCES

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